CS 1103

BIRD SPECIES DATABASE PROJECT

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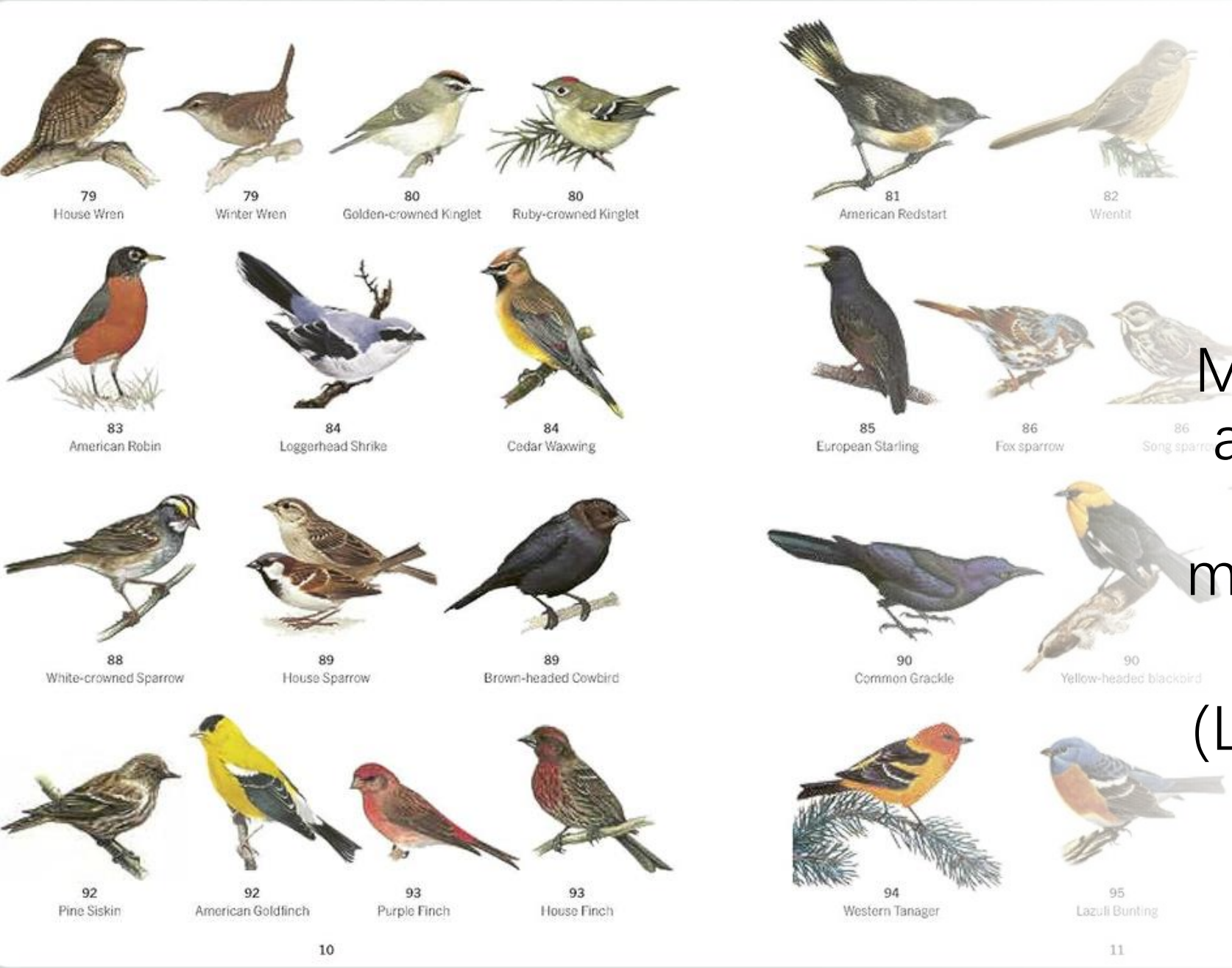
March 24th, 2026



Project Overview

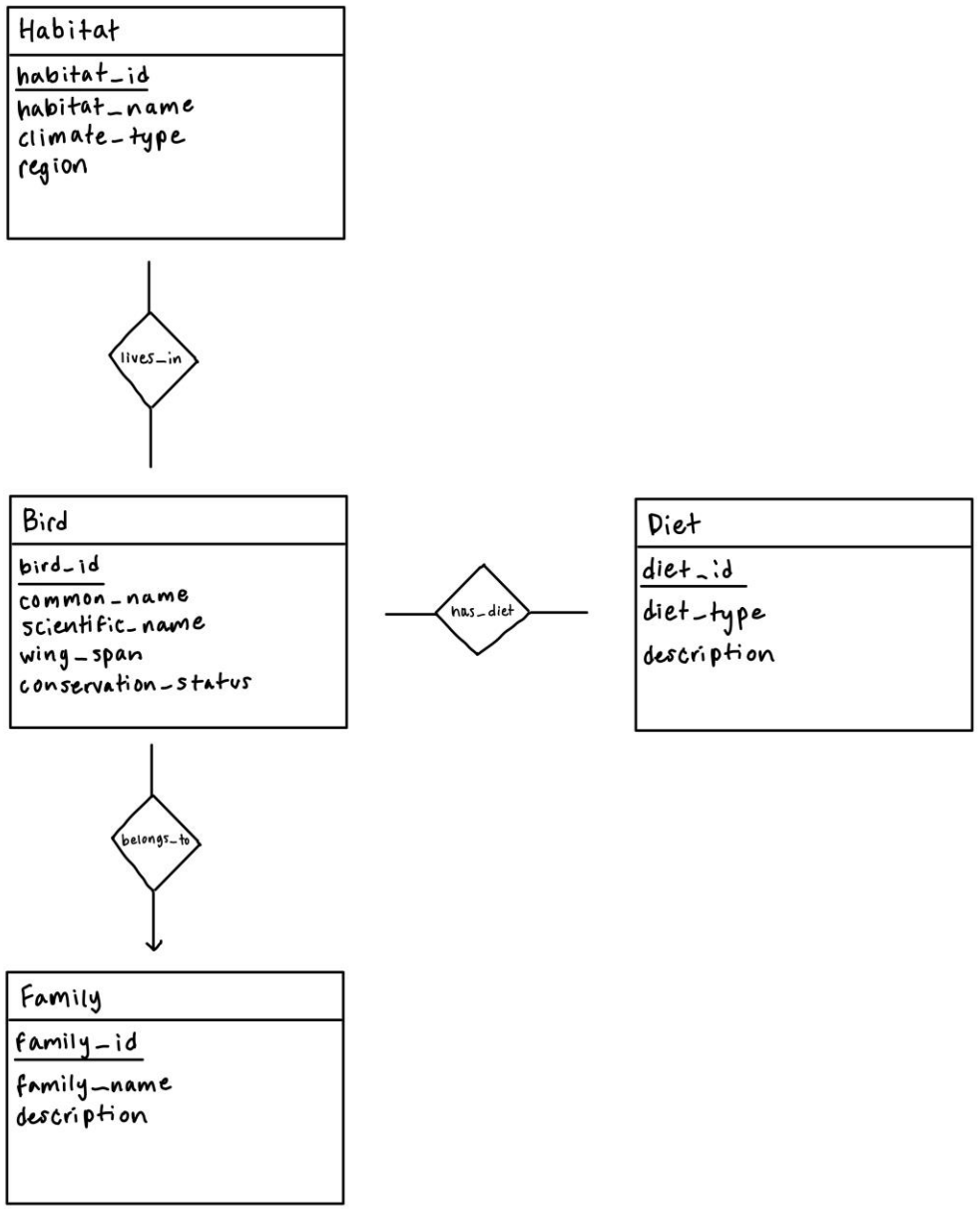
My goal is to create a database system to store and manage information about birds.

(Like the app Merlin Bird ID)

MINIMAL REQUIREMENTS

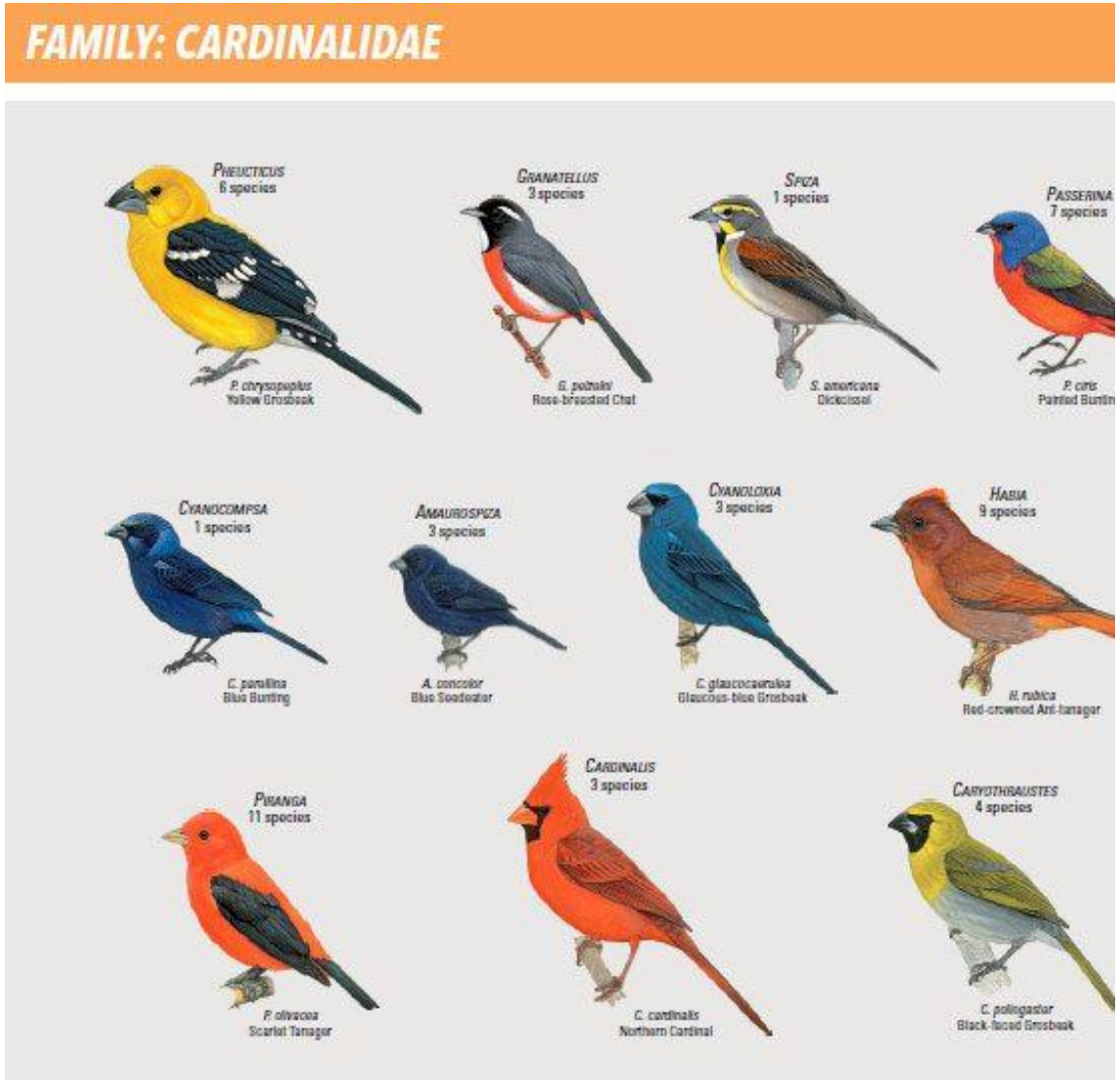
ER DIAGRAM

shows the main entities and their relationships



Relationships

- Every bird belongs to one family (one-to-many)
- Every bird lives in habitat or multiple habitats (many-to-many)
- Every bird has a diet or multiple diets (many-to-many)



DDL statements for each entity

```
CREATE TABLE Bird (
    species_id INT PRIMARY KEY,
    common_name VARCHAR(100),
    scientific_name VARCHAR(100),
    wing_span DOUBLE,
    conservation_status VARCHAR(50),
    family_id int,
    FOREIGN KEY (family_id)
    REFERENCES Family(family_id)
);

CREATE TABLE Diet (
    diet_id INT PRIMARY KEY,
    diet_type VARCHAR(100),
    description VARCHAR(225)
);
```

```
CREATE TABLE Family (
    family_id INT PRIMARY KEY,
    family_name
    VARCHAR(100),
    description VARCHAR(225)
);

CREATE TABLE Habitat (
    habitat_id INT PRIMARY KEY,
    habitat_name VARCHAR(100),
    climate_type VARCHAR(100),
    region VARCHAR(100)
);
```

DDL statements for each relationship

```
CREATE TABLE BirdHabitat (
    bird_id INT,
    habitat_id INT,
    PRIMARY KEY (bird_id, habitat_id),
    FOREIGN KEY (bird_id) REFERENCES Bird(bird_id),
    FOREIGN KEY (habitat_id) REFERENCES Habitat(habitat_id)
);
CREATE TABLE BirdDiet (
    bird_id INT,
    diet_id INT,
    PRIMARY KEY (bird_id, diet_id),
    FOREIGN KEY (bird_id) REFERENCES Bird(bird_id),
    FOREIGN KEY (diet_id) REFERENCES Diet(diet_id)
);
```

Sample insert statements for each table

- INSERT INTO Bird VALUES (1, 'BaldEagle', 'Haliaeetus leucocephalus','Accipitridae', 'Least Concern');

- INSERT INTO Habitat VALUES (1, 'Forest','Temperate', 'North America');

- INSERT INTO Family VALUES (1,'Anseriformes', 'Waterfowl & Relatives');

- INSERT INTO Diet VALUES (2, 'Frugivorous', 'Primarily consumes fruit);

Sample insert statements for each table

- INSERT INTO BirdHabitat VALUES (1, 1);
- INSERT INTO BirdDiet VALUES(1, 1);

Connect to the database using JDBC

I was able to connect to the database using JDBC with this code (lab 7):

```
Connection conn =
DriverManager.getConnection
("jdbc:sqlite:bird.db") ;
```

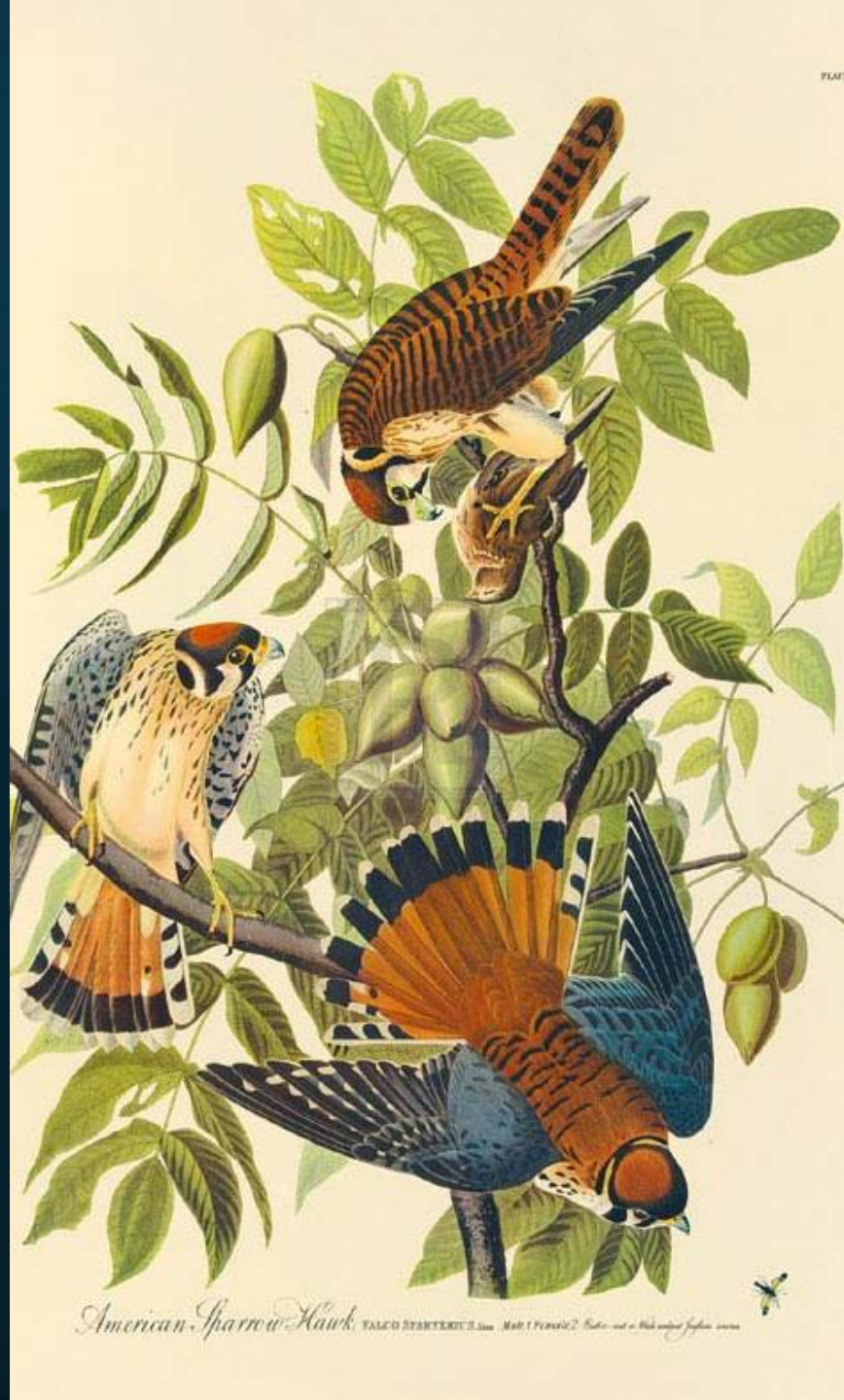
This creates a database file titled "bird.db"

Create database schema using JDBC

- Used JDBC to create tables with SQL
- Created tables based on ER diagram

```
import java.sql.*;

public class BirdDatabase {
    public static void main(String[] args) {
        try {
            //connect to SQLite database
            Connection conn = DriverManager.getConnection("jdbc:sqlite:bird.db"
            Statement stmt = conn.createStatement();

            System.out.println("Connected to database.\n");

            //create tables (schema)

            stmt.executeUpdate("CREATE TABLE IF NOT EXISTS Family (" +
                    "family_id INTEGER PRIMARY KEY, " +
                    "family_name TEXT)");

            stmt.executeUpdate("CREATE TABLE IF NOT EXISTS Bird (" +
                    "bird_id INTEGER PRIMARY KEY, " +
                    "common_name TEXT, " +
                    "scientific_name TEXT, " +
                    "conservation_status TEXT, " +
                    "family_id INTEGER, " +
                    "FOREIGN KEY (family_id) REFERENCES Family(family_id))");
```

MODERATE REQUIREMENTS



Create database instance

Inserted some bird and family data into the database:

```
stmt.executeUpdate("INSERT OR IGNORE INTO Bird VALUES (1, 'Northern Cardinal', 'Cardinalis cardinalis', 'Least Concern', 1)");

stmt.executeUpdate("INSERT OR IGNORE INTO Bird VALUES (2, 'Bohemian Waxwing', 'Bombycilla garrulus', 'Least Concern', 2)");
```

SQL statements for select, update, and delete

```
SELECT b.common_name, f.family_name
FROM Bird b
JOIN Family f ON b.family_id = f.family_id;

DELETE FROM Bird
WHERE bird_id = 2;

UPDATE Bird
SET conservation_status = 'Endangered'
WHERE bird_id = 1;
```

Output

```
C:\Windows\System32\cmd.e

Tables created.

Sample data inserted.

All Birds:
1 | Northern Cardinal | Cardinalis cardinalis
2 | Bohemian Waxwing | Bombycilla garrulus
3 | Great Blue Heron | Ardea herodias
4 | Blue Jay | Cyanocitta cristata
5 | Common Raven | Corvus corax
6 | American Goldfinch | Spinus tristis
7 | Canada Goose | Branta canadensis

Inserted Snowy Owl.
Updated Northern Cardinal status.
Deleted Snowy Owl.

Final Bird List:
1 | Northern Cardinal | Endangered
2 | Bohemian Waxwing | Least Concern
3 | Great Blue Heron | Least Concern
4 | Blue Jay | Least Concern
5 | Common Raven | Least Concern
6 | American Goldfinch | Least Concern
7 | Canada Goose | Least Concern

Done.
```